



Investigating transformations with Desmos

Task A: Investigating translation with Desmos

Draw any polygon and translate it to the right using a vector.

- 1) Click on your object and drag it around the workspace.

What happens to the image? Why does it do this?

The image moves with the object. This is because the vector is the same, so the image is always to the right of the object and the same distance away.

- 2) Click on the end of your vector and drag to make the vector longer.

What happens to the image? Why does it do this?

The image moves further away from the object. This is because a longer vector means the movement is a greater distance.

- 3) Click on the middle of your vector and drag it around the workspace.

What happens to the image? Why does it do this?

The image does not move. This is because the vector still has the same size and direction.

Task B: Investigating rotation with Desmos

Draw any polygon, rotate it using a slider and set the slider to 270° .

- 1) Drag the centre of rotation towards the image.

What happens to the image? Why does it do this?

The image moves away. This is because the centre of rotation has moved further away from the object, so the distance to the image increases too.

- 2) Is it possible for the centre of rotation to be inside the image?

If so, how? If not, why not?

Yes. When the centre of rotation is inside the object, it is also inside the image.

- 3) Move the spot in the slider left and right. For what angles does the image perfectly overlay the object? Why does it do this for these angles?

0° because the image has not turned. 360° because its a full turn.

If the slider went further, then it would also overlay perfectly for any multiple of 360° .



Task C: Investigating reflection with Desmos

Draw a polygon where every edge is a different length (e.g. a scalene triangle).

Draw a vertical line and reflect the polygon over the line.

1) Drag the line of symmetry towards the image.

What happens to the image? Why does it do this?

The image moves away. This is because the line of symmetry is always equidistant between the object and image.

2) Is it possible for the line of symmetry to be inside the image?

If so, how? If not, why not?

Yes. When the line of symmetry is inside the object.

3) Is it possible for the image to perfectly overlay the object?

If so, how? If not, why not?

No. This is because the image does not retain the sense of the object.

Task D: Investigating enlargement with Desmos

Draw any polygon and enlarge it with a slider scale factor.

1) Drag the slider left and right.

a) When is the image larger than the object? *scale factor > 1*

b) When is the image the same size as the object? *scale factor $= 1$*

c) When is the image smaller than the object? *scale factor < 1*

d) What happens to the image when the scale factor is 0? Why does this happen?

The image disappears because it has no size.

2) Select the line tool.

Click on a vertex on the object and its corresponding vertex on the image to draw a line that traverses both vertices. Repeat for all pairs of corresponding vertices.

Where do the lines intersect each other?

At the centre of enlargement.

